Summary of GenAI Use – ChatGPT

Throughout this conversation, I have assisted the student with understanding and completing parts of INF201 Deliverable 1.

My assistance has mainly focused on explaining the different tasks, discussing possible approaches, and helping the student understand how to work with weather data from the Norwegian Meteorological Institute's Frost API.

I have helped the student with the following:

1. **Understanding the tasks:** I explained the requirements of Tasks 1, 2, 3, and 5, including how to retrieve weather data, create plots, calculate summary statistics, and understand moving averages.

2. **Discussing code structure:** I reviewed the student's ideas for organizing the code and provided feedback on different approaches. We discussed how to structure functions, organize data using pandas, and keep the code simple and readable.

3. **Code examples and explanations:** I provided Python examples based on the methods demonstrated in the official Frost API Python example (https://frost.met.no/python_example.html). I explained how API requests work, how to extract observations, and how to use libraries such as requests, pandas, and matplotlib.

4. **Code review and improvements:** I reviewed code shared by the student and suggested possible improvements. These included simplifying functions, improving plots, checking for missing observations, and making summary tables easier to create. I also explained why these changes could improve readability, reliability, or the presentation of the results.

5. **Troubleshooting and development tools:** I helped identify and resolve errors involving Jupyter Notebook, API authentication, missing data, and Python variables. I also provided guidance on using Git branches, committing changes, and pushing notebooks to GitHub.

Throughout the discussion, I explained the reasoning behind suggested changes and provided alternative solutions when the student wanted simpler code. The student was able to test the examples, review the results, and discuss further adjustments.

Overall, ChatGPT was used as a learning resource, programming assistant, and code review tool to support the student's understanding of Python, data processing, and visualization.